Lesson 6 Teacher's Key

Scratch

INSTRUCTOR ONLY - do not distribute to students

Level 1 - Complete Worked Solution

A dedicated GameManager sprite owns all win-tracking, listening for a broadcast from the Enemy's existing clone script:

```
-- Enemy sprite (spawn script, unchanged from Lesson 5):
when green flag clicked
  set [enemy_number] to 0
  repeat 5
    change [enemy_number] by 1
    set [enemy_x] to ((enemy_number * 60) - 150)
    create clone of [myself]

-- Enemy sprite (clone script):
when I start as a clone
  show
  go to x: (enemy_x) y: 150
  forever
    if <touching [Bullet]?> then
      broadcast [enemy hit]
      delete this clone

-- GameManager sprite (new, hidden):
when green flag clicked
  hide
  set [enemies_remaining] to 5
when I receive [enemy hit]
  change [enemies_remaining] by -1
  if <enemies_remaining = 0> then
    broadcast [you win]

-- Player sprite (new script):
when I receive [you win]
  say [You Win!] for 3 seconds
```

Level 2 - Complete Worked Solution

The decrement and the win check both live inside GameManager's single "when I receive [enemy hit]" script, which runs once per broadcast, and each Enemy clone broadcasts exactly once, right on its own hit - there's no separate code path that could double-count or skip an enemy, so "You Win!" fires exactly once, exactly on the last hit.

Level 3 - All Worked Solutions

Every stretch-goal idea from the handout, fully worked - not just one. Useful for validating each idea actually works before class, and as a ready reference no matter which one a student ends up picking.

Sound: celebration sound

Add the provided sfx_win.wav right before the say block on the Player's win script:

```
when I receive [you win]
  start sound [sfx_win]
  say [You Win!] for 3 seconds
```

Visual: bigger celebration

A couple of size pulses read more like a celebration than a plain message:

```
when I receive [you win]
  set size to 120%
  wait 0.1 seconds
  set size to 100%
  say [You Win!] for 3 seconds
```

Structure: one place tracks the win

The decrement, the zero-check, and the broadcast all stay inside GameManager's own script - the Enemy sprite only ever broadcasts that it was hit, and no second script anywhere else also reads or writes enemies_remaining.

Complete Level 3 solution (all 2 goals combined)

The win-tracking logic unchanged, plus the Player's enhanced win script:

```
-- Enemy sprite (clone script, unchanged):
when I start as a clone
  show
  go to x: (enemy_x) y: 150
  forever
    if <touching [Bullet]?> then
      broadcast [enemy hit]
      delete this clone

-- GameManager sprite (unchanged):
when green flag clicked
  hide
  set [enemies_remaining] to 5
when I receive [enemy hit]
  change [enemies_remaining] by -1
  if <enemies_remaining = 0> then
    broadcast [you win]

-- Player sprite:
when I receive [you win]
  start sound [sfx_win]
  set size to 120%
  wait 0.1 seconds
  set size to 100%
  say [You Win!] for 3 seconds
```